C05 Physics 2D

CRE132 worksheet

What you'll make

A side-view physics playground: crates, rocks, pots and a log that fall, roll, bounce and slide, and a player who walks into them and pushes them about. This time the player can't walk through things.

You'll learn the jobs of a **Rigidbody 2D** (it makes things move under physics) and a **Collider 2D** (it gives them a shape), the three **body types**, **Gravity Scale**, **Physics Materials** for friction and bounce, and how a whole tilemap becomes one smooth collider. Then you'll rewrite C04's movement so it goes through the Rigidbody 2D.

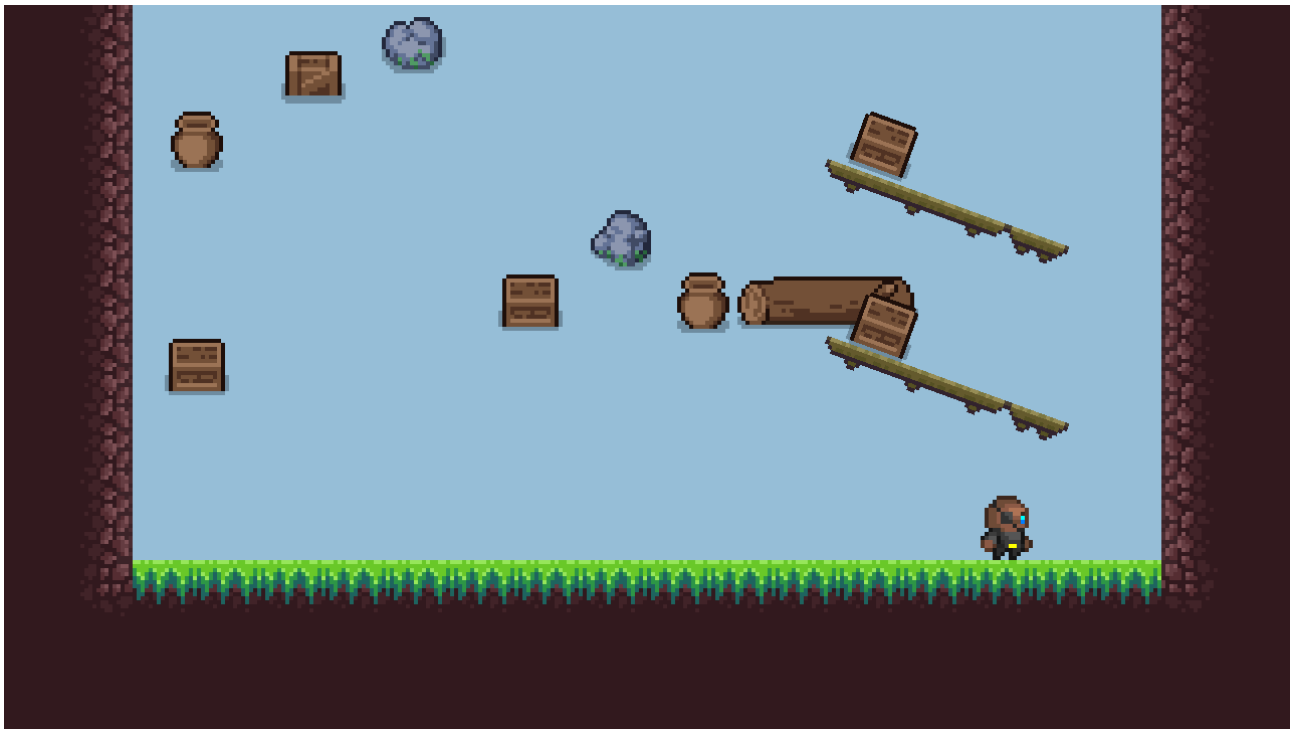


Figure 1: The finished scene: props falling and settling between two cliffs, with two sloping planks and the player on the ground.

Before you start

1. Unzip `CRE132_C05_Physics2D_Starter.zip`. In Unity Hub, choose **Add > Add project from disk** and pick the `CRE132_C05_Physics2D_Starter` folder. Open it with Unity **6000.3.11f1**.
2. Open the scene `Assets/_game/Scenes/Core/C05_Physics2D`.
3. Press **Play**. The props in **Body Types** and **Collider Shapes** fall and settle: their physics is already set up. The props in **Gravity And Bounce** and **Friction** hang in the air without moving: they have no physics yet. The player drops onto the ground but can't move. Press Play again to stop.

The scene has a **Grid** with the **Ground** tilemap, four groups of props (**Body Types**, **Collider Shapes**, **Gravity And Bounce**, **Friction**) and the **Player**. You'll look at how the first two groups work, add a prop of your own, give the other two groups their physics, then write the player's script.

Making a script: in the Project window, go to the folder, right-click and choose **Create > Scripting > MonoBehaviour Script**. Type the name straight away and press Enter. The name must match the class name in the code exactly, capital letters included.

1. Rigidbody 2D and body types

A **Rigidbody 2D** hands an object over to the physics engine: gravity, pushes and collisions move it. Open **Body Types** and click its two objects:

- **Dynamic Pot:** Body Type **Dynamic**. Gravity pulls it and things can push it.
- **Kinematic Crate:** Body Type **Kinematic**. It ignores gravity and forces and is only moved by code, so it hangs in the air and catches the pot.
- The third type, **Static**, never moves at all.

Add it: make your own crate. Open `Assets/_game/Art/Sprites/MysticWoods`, click the arrow on `objects.png` and drag the `crate` sprite into the Scene view, up in the air. Name it **My Crate**. **Add Component > Rigidbody 2D** (leave it **Dynamic**).

Test it: press Play. Your crate falls… straight through the ground. A Rigidbody 2D moves things, but without a **collider** they have no shape to bump into.

2. Colliders

Add it: on **My Crate**, **Add Component > Box Collider 2D**. Unity sizes the box to the sprite.

Test it: press Play. The crate lands on the ground. If you can't see the green collider outlines in the Scene view, turn on **Gizmos** (the button at the top right of the Scene view).

Now open **Collider Shapes** and compare the four props as they land:

Object	Collider
Crate (Box)	Box Collider 2D
Rock (Circle)	Circle Collider 2D: rolls
Pot (Polygon)	Polygon Collider 2D, traced from the sprite
Log (Capsule)	Capsule Collider 2D: rounded ends

3. Static colliders and the tilemap collider

A collider without a Rigidbody 2D is *static*: things collide with it, but it never moves. That's ideal for scenery. Open **Friction**: it has two sloping planks, each with a crate resting on it.

Add it:

1. Click **Ice Plank** and choose **Add Component > Box Collider 2D**. Unity sizes the box to the plank (**Size** 4 × 0.375) and tilts it with the plank. Don't add a Rigidbody 2D.
2. Do the same for **Grippy Plank**.
3. Click **Crate On Ice Plank** and add a **Rigidbody 2D** (leave it **Dynamic**) and a **Box Collider 2D**.
4. Do the same for **Crate On Grippy Plank**.

Test it: press Play. Both crates stay on their planks instead of falling through them, and the planks stay where they are.

Now click **Ground** (inside **Grid**). It has three components working together:

- **Tilemap Collider 2D** gives every tile its own box.
- **Composite Collider 2D** joins those boxes into one shape (**Geometry Type: Polygons**). The Tilemap Collider's **Composite Operation** is set to **Merge** so its boxes feed into it.

- **Rigidbody 2D**, Body Type **Static**. The Composite Collider needs a Rigidbody 2D; Static means the ground never moves.

See why: set the Tilemap Collider 2D's **Composite Operation** to **None** and look at the outlines in the Scene view. Every tile is a separate box, and a sliding crate can catch on the joins between them. Set it back to **Merge**.

4. Gravity Scale

Open **Gravity And Bounce**. It has two props, still without physics.

Add it:

1. Click **Floaty Box (Gravity 0.2)**. Add a **Rigidbody 2D** and set its **Gravity Scale** to **0.2**, so it falls at a fifth of normal gravity. Add a **Box Collider 2D**.
2. Click **Bouncy Rock**. Add a **Rigidbody 2D** (leave Gravity Scale at **1**) and a **Circle Collider 2D**, so it can roll.

Test it: press Play. The rock drops and lands without bouncing. The box drifts down slowly.

Now try it on your own crate: set **My Crate**'s Gravity Scale to 0.2, then 0, then −0.5, pressing Play each time. Set it back to 1. Gravity for the whole project is in **Edit > Project Settings > Physics 2D** (Y **−9.81**).

5. Physics Materials

A **Physics Material 2D** sets how slippery (**Friction**) and bouncy (**Bounciness**) a collider is. Without one, a collider uses Unity's default: Friction 0.4, Bounciness 0.

Add it:

1. In the Project window, open `Assets/_game/Materials`. Right-click, choose **Create > Folder** and call it `Physics`.
2. Open `Physics`, right-click and choose **Create > 2D > Physics Material 2D**. Name it `Bouncy`. In the Inspector, set **Friction** to **0.4** and **Bounciness** to **0.8**.
3. Make `Ice` and `Grippy` the same way, with these values:

Material	Friction	Bounciness	Goes on
Bouncy	0.4	0.8	Bouncy Rock
Ice	0	0	Ice Plank
Grippy	1	0	Grippy Plank

4. Click **Bouncy Rock** and drag `Bouncy` from the Project window into the **Material** slot of its **Circle Collider 2D**.
5. Drag `Ice` into the **Material** slot of **Ice Plank**'s Box Collider 2D, and `Grippy` into **Grippy Plank**'s.

Test it: press Play. The rock bounces several times before it settles. Watch **Friction**: the two planks have the same slope, but the crate on Ice slides off and the one on Grippy stays put.

6. Moving the player with physics: PhysicsPlayerMovement

The **Player** already has a **Rigidbody 2D** (Dynamic) and a **Capsule Collider 2D**. C04's script moved `transform.position`, which ignores physics. This one sets the Rigidbody 2D's velocity instead.

Make a folder `Assets/_game/Scripts/C05_Physics2D` and create `PhysicsPlayerMovement` in it:

```
// Moves the player left and right with a Rigidbody2D instead of changing
// transform.position.
// Because physics moves the player, it now stops at walls, stands on the ground and
// can push crates.
using UnityEngine;

public class PhysicsPlayerMovement : MonoBehaviour
{
    [SerializeField] private float moveSpeed = 4f;

    private Rigidbody2D body;
    private float horizontalInput;

    private void Awake()
    {
        // GetComponent finds the Rigidbody2D on this same GameObject.
        body = GetComponent<Rigidbody2D>();
    }

    private void Update()
    {
        // Read input in Update, which runs every frame, so no key press is missed.
        horizontalInput = Input.GetAxis("Horizontal");
    }

    private void FixedUpdate()
    {
        // Physics runs in FixedUpdate, so we change the Rigidbody2D here.
        // We set the sideways speed and keep the current vertical speed, so gravity
        // still works.
        // Velocity is already "units per second", so there is no Time.deltaTime here.
        body.linearVelocity = new Vector2(horizontalInput * moveSpeed,
            body.linearVelocity.y);
    }
}
```

What's going on:

- **`GetComponent<Rigidbody2D>()`** finds the Rigidbody 2D on the same object. `Awake` runs once, before `Start`, so `body` is ready before anything uses it.
- **`Update` reads the input, `FixedUpdate` moves.** `FixedUpdate` runs in step with the physics engine (50 times a second), which is where Rigidbody changes belong. Reading keys in `Update`, every frame, means no press is missed. The `horizontalInput` variable carries the value from one to the other.
- **`linearVelocity`** is the speed and direction the body is moving. We set the sideways part (`x`) and **keep the current `y`**, so gravity can still pull the player down.
- **No `Time.deltaTime`:** velocity is already in units per second.

Add it: select the **Player**, **Add Component > PhysicsPlayerMovement**. Leave **Move Speed** at **4**.

Test it: press Play and walk with **A/D** or the **arrows**. The player stands on the ground, stops at the cliffs and pushes crates.

7. Freeze Rotation

On the Player's Rigidbody 2D, open **Constraints**: **Freeze Rotation Z** is ticked. Untick it, press Play and walk into a crate: the player tips over like any other box. Tick it again.

Check your work

- ☐ Your crate falls and lands on the ground (not through it).
- ☐ The Floaty Box drifts down slowly and the Bouncy Rock bounces before it settles.
- ☐ `Assets/_game/Materials/Physics` has `Bouncy`, `Ice` and `Grippy`, each on its collider.
- ☐ The crate on the Ice plank slides off; the one on the Grippy plank stays.
- ☐ A/D or the arrows walk the player; it stops at the cliffs and pushes props.
- ☐ The player stays upright.
- ☐ Compare with `CRE132_C05_Physics2D_Finished` if something behaves differently.

Key points

- **Rigidbody 2D moves, Collider 2D gives a shape.** You need both for an object to fall and land. A collider on its own is static scenery.
- **Move Rigidbody objects through the Rigidbody** (velocity or forces), not `transform.position`, or physics can't stop them at walls.
- **`Update` for input, `FixedUpdate` for physics.** Velocity is already per second, so no `Time.deltaTime`.
- **A Composite Collider 2D** turns a tilemap's boxes into one smooth shape.

If something goes wrong

- **Things fall through the ground or a plank:** the falling object has no collider, or the plank has none. If it's the ground, check Composite Operation is **Merge**.
- **A plank falls when I press Play:** you gave it a Rigidbody 2D. Remove it (the three dots on the component, **Remove Component**): scenery only needs a collider.
- **The Ice crate doesn't slide:** the `Ice` material is on the crate or on nothing, not in the **Material** slot of **Ice Plank**'s Box Collider 2D.
- **NullReferenceException in PhysicsPlayerMovement:** the Player has no **Rigidbody 2D**.
- **The player doesn't move:** the script isn't on the Player, or Move Speed is 0.
- **The player falls over: Freeze Rotation Z** is unticked.
- **The player stops at a crate instead of pushing it:** that crate is Kinematic or has no Rigidbody 2D. Only Dynamic bodies get pushed.

Try it

1. Set the player's **Gravity Scale** to 0 and use `Input.GetAxis("Vertical")` for the `y` velocity: you have C04's top-down movement, but now walls stop you.
2. Give the player's Capsule Collider 2D the **Ice** material. What happens when you let go of the key? Then think about why it doesn't slide anyway. (Hint: what does `FixedUpdate` set the velocity to when no key is held?)
3. Change the Ground's Composite Collider **Geometry Type** to **Outlines** and drop a crate so it starts inside the ground. What's the difference?
4. Make a material `MyBouncy` with **Bounciness** 1 and put it on **My Crate**'s Box Collider 2D. How long does your crate keep bouncing?